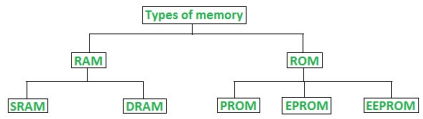
What Is RAM ?

14 July 2024 07:45



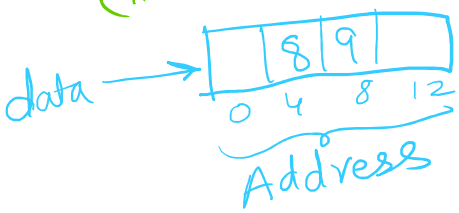
Classification of computer memory

Random Access Memory (RAM) is a type of computer memory that is used to temporarily store data that the computer is currently using or processing. RAM is volatile memory, which means that the data stored in it is lost when the power is turned off. RAM is typically used to store the operating system, application programs, and data that the computer is currently using.

Read Only Memory (ROM) is a type of computer memory that is used to permanently store data that does not need to be modified. ROM is non-volatile memory, which means that the data stored in it is retained even when the power is turned off. ROM is typically used to store the computer's BIOS (basic input/output system), which contains the instructions for booting the computer, as well as firmware for other hardware devices.

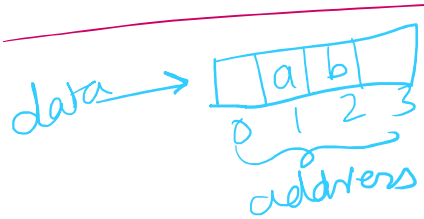
RAM = Random Access Memory

(Random allocation of data in RAM)



0, 4, 8, 12 } address of int, float, double,

×4



0, 1, 2, 3 } address of Char → ASCII

+1

1 bit → 0 | 1

1 byte → 8 bits

1024 bytes → 1 KiloByte

1024 KiloByte → 1 MegaByte

⋮

A 64-bit processor refers to a microprocessor that can process data and instructions in chunks of 64 bits. Microprocessors that can handle 64 bits perform a larger number of calculations per second compared to 32-bit processors.

×64 → 64 bit computers

×32 → 32 bit computers

} memory allocation

** We don't know where RAM will store the data.

1 in 32 bit Computer

$$\underbrace{0000 \ldots\ldots 0}_{31\ 0's}1$$

that's how it works!!!